DC Performance Results Versus Assessment of ITER Main Busbar NbTi Conductors

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***Abstract*—Four ITER main busbar (MB) conductor samples were tested in the SULTAN test facility (Centre de Recherches en Physique des Plasma, Ecole Polytechnique Fédérale de Lausanne, Switzerland) between 2011 and 2013. The MB conductors are NbTi-based cable-in-conduit conductors (CICCs), and they will become part of the feeder system of the ITER magnets. The measured dc performance of the four MB samples varied significantly, supposedly depending on the design of the bottom terminations. Two out of three samples with a U-bend box, made of a continuous conductor section, exhibit approximately 0.5 K lower current-sharing temperature T_{cs} than the sample with a solder-filled bottom joint, consisting of two straight conductor sections. We assess the theoretically expected T_{cs} performance of the MB conductor based on the characterization of individual NbTi strands, on the ITER NbTi scaling law, and on the magnetic field distribution across the cable cross section. The magnetic field in a SULTAN sample consists of three components, namely, the background SULTAN field, the self-field generated by the current in the conductor under test, and the magnetic field generated by the current in the return conductor. Taking into account all the three components, we calculate the average electric field in the cable and determine T_{cs} as in the experiment, namely, as the temperature at which the electric field reaches the critical value of $E_c = 0.1$ μV/cm. The theoretically assessed T_{cs} confirms that the MB sample with solder-filled joint behaves as expected.**

***Index Terms*—Cable-in-conduit conductor (CICC), current sharing temperature, ITER, NbTi.**

I. Introduction

THE main busbar conductors (MB) in ITER magnet system provide electrical as well as cryogenic connection of Toroidal Field (TF), Central Solenoid (CS), and Poloidal Field (PF) coils to the current leads. The MB conductors are based on Cable-In-Conduit Conductor (CICC) technology. The cable has round shape, with a copper-wire core in the center surrounded with multiple stage NbTi sub-cables. Main characteristics of the MB conductors are summarized in Table I, more details can be found e.g. in [1]. The NbTi strands for MB conductors were manufactured by the Western Superconducting Technologies Company (WST) in China [2].

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TABLE I
MAIN PARAMETERS OF THE MB CONDUCTORS

Characteristics	value
Outer diameter	44.5 mm
Total number of strands	1483
Number of supercond. strands	900
Number of 0.73 mm Cu strands	522
Number of 2 mm Cu wires (inner bundle)	61

TABLE II
T_{cs} MEASURED IN SULTAN FOR MB SAMPLES AT THE BACKGROUND FIELD OF 3.22 T AND AT THE CURRENT OF 45.5 kA. ITER ACCEPTANCE STANDARD IS 6.7 K

Sample	MBCN1	MBCN2	MBCN3	MBCN4
Tcs(3.22T,45.5kA)	6.95 K	6.51 K	6.45 K	7.02 K

II. Background

Four MB samples have been manufactured in China, under the supervision of China Domestic Agency (CDA) and of the Institute of Plasma Physics of the Chinese Academy of Sciences (ASIPP), and subsequently tested in the SULTAN facility [3] at CRPP, EPFL in Switzerland. The main purpose of the tests was the quality control [4] of the produced MB conductors, namely the measurement of the current sharing temperature, T_{cs}. The details about the testing of MB conductors in SULTAN can be found in [1].

In SULTAN, the performance of a superconducting sample is measured in a high-field region, which is ~45 cm long. The first MB sample, MBCN1, was tested in 2011. The performance of the sample was satisfactory [1], the MBCN1 performance fulfilled the ITER requirements. On contrary, the performance of the second MB sample, MBCN2, tested in 2013, was significantly worse than that of MBCN1, see Table II. The degraded performance of MBCN2 could be explained by a premature quench initiation in the U-bend box, while the MB conductor itself could have had better performance than observed in the measurement. The bending radius of the individual sub-cables in the U-bend can be quite small, as shown in Fig. 1, leading to a big self-field generated in some U-bend locations. In this case, the sum of the background field and self-field in the U-bend location can exceed the field in the high-field region.

The hypothesis that the quench was initiated in the U-bend, rather than in the SULTAN test region, is supported by a faster temperature increase in front of the high-field region (upstream) than behind it (downstream) during the quench initialization. It is interesting to note that also the MBCN1 sample was made with the U-bend termination, however its performance was not

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Fig. 1. (Top) U-bend box of the MBCN3 sample and (bottom) the solder-filled bottom joint similar to that done in the MBCN4 sample.

affected. In addition, the PF conductor samples, which are also produced with a U-bend box, do not seem to suffer from the same problem. In the PF samples, however, the background SULTAN field is typically higher and the operating current smaller than in the MB samples, and therefore the self-field effect plays a smaller role.

The MBCN2 sample was subsequently modified at ASIPP. The copper-core bundle was removed in the U-bend region in order to have better control over the individual sub-cable bending path. In addition, the width of the U-shape box was increased from 128 mm to 136 mm, and the central steel stiffener was prolonged by 25 mm in order to better control the conductor bending. The modified sample, called MBCN3, was retested in SULTAN, unfortunately without any improvement of measured $T_{\rm cs}$.

In order to avoid the suspected bad effect of the sub-cable bending causing the premature quench initiation in the U-bend, a solder-filled bottom joint has been employed in the next MBCN4 sample (Fig. 1). In this case, only the conductor sections were fabricated in China, while the joint preparation, assembly and instrumentation was done at CRPP.

The solder-filled joint preparation followed, wherever applicable, the joint preparation procedure for the TF SULTAN samples [5]. The outer conductor jacket as well as the inner copper bundle have been cut out in the termination region, Fig. 2. The copper bundle was replaced by a steel pipe, temporarily filled with a Teflon rod. Subsequently, the strand nickel coating was removed by nitric acid in an ultrasound bath, and the terminations were filled with solder. In the end, the Teflon rods were removed such that the helium can pass through the steel pipes.

Due to different geometry, the self-field generated in the solder-filled joint is smaller than in the U-bend box. $T_{\rm cs}$ measured on the MBCN4 sample was very close to the MBCN1 sample, i.e., approx. 0.5 K above the $T_{\rm cs}$ of MBCN2 and MBCN3 (at 3.22 T and 45.5 kA), see Table II. In this paper

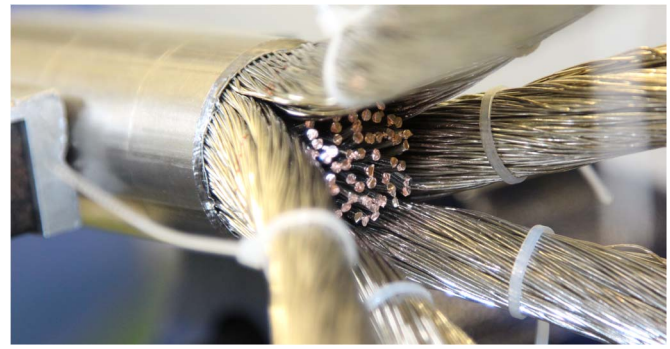

Fig. 2. MBCN4 conductor leg after removal of the inner copper core.

TABLE III
PARAMETERIZATION OF THE WST NbTi STRANDS ADAPTED FROM [2]

B_{c20}(T)	C_0(Nm^{-3})	T_{c0} (K)	α	β	γ	n
13.72	113 200	8.79	0.98	0.98	1.96	1.70

we attempt to assess the theoretically expected value of $T_{\rm cs}$, and to compare it with the measured MB performance. We are interested to see, whether the performance of MBCN1 and MBCN4 corresponds to the expectation based on the individual strand properties.

Discussions of the MBCN2 and MBCN3 degraded performance raised the question, how does the MB performance depend on the cable compaction. On request of ITER Organization, the left conductor section for MBCN4 was compacted below the MB nominal outer diameter of 44.5 mm down to 43.3 mm, corresponding to the void fraction of 33% instead of nominal 38%. The motivation to try the bigger compaction was to reduce the inter-strand resistance, and consequently to improve the current redistribution within the tested conductor, if the current imbalance in the sample was responsible for the premature quenches in MB samples.

III. Theoretical Assessment

Our theoretical assessment of the MB conductor is strictly based on the NbTi strand properties, as they were measured and reported in [2], and on the geometry of the MB conductor/sample. There is no free parameter used in our assessment, e.g., we had no possibility to "tune" our theoretical prediction to match the measurement. The only fit we did concerned the n-value dependence on critical current density, J_c, as will be explained later. The $n(J_c)$ dependence belongs to the strand characterization.

A. Strand Properties and Scaling Law

In order to scale critical current density to a given magnetic field and temperature, we use the Bottura scaling law for NbTi strands [6], which is the scaling law officially adopted by ITER. The scaling-law equation defining the critical current reads:

$$Jc = C_0/B \cdot b^{\alpha}(1-b)^{\beta}(1-t^n)^{\gamma} \quad (1)$$

where b is the reduced field and t the reduced temperature defined in [6]. The numerical values of scaling law parameters for WST strands, reported in [2], are summarized in Table III.

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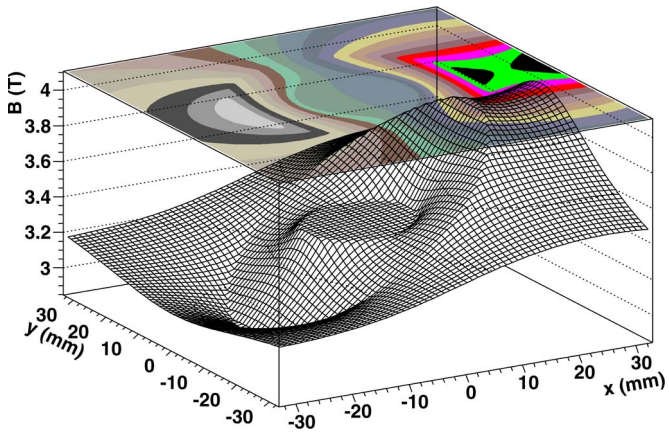

Fig. 3. Magnitude of the magnetic field perpendicular to a conductor leg. Only the field profile across one conductor leg and its neighboring region is shown for the SULTAN background field of 3.22 T and transport current of 45.5 kA. The distance between the centers of the two sample legs is 51 mm.

In addition to the scaling law, we also need to know the dependence of the n-value on the critical current density, J_c. The measurement of $n(J_c)$ was done for the WST strands at the WST company for critical current densities up to 5000 A/mm^2. The measurement can be well approximated by the polynomial fit

$$n(J_c) = 5.26 + 0.0226 \cdot J_c - 4.657 \cdot 10^{-6} \cdot J_c^2 + 3.319 \cdot 10^{-10} \cdot J_c^3 \quad (2)$$

where J_c is in A/mm^2.

B. Magnetic Field in the MB Samples in SULTAN

The magnetic field sensed by the strands inside the MB samples consists of three components: (1) the SULTAN background field, which is more-or-less constant (homogeneity 1%) within the tested conductor region, (2) the self-field generated by the current transmitted through the strands of the conductor under test, and (3) the return cable field, generated by the current in the other conductor section. The superposition of the three components is discussed in detail in [7]. Here, in Fig. 3, we just illustrate the resultant B field magnitude across one MB conductor in the SULTAN background field of 3.22 T and transport current of 45.5 kA. The region of (almost) constant field in the conductor center corresponds to the central copper bundle, which does not transport any current in the superconducting state.

C. $T_{\rm cs}$ Prediction

The easiest and straightforward way of $T_{\rm cs}$ assessment in NbTi cables is to find out the temperature, at which the critical current defined in (1) equals to the operating current. The peak field, i.e. the highest field found within the cable cross section in Fig. 3, is typically used in this approach, which we will call "simple approach" in the following discussion.

In this work, we prefer to use more sophisticated $T_{\rm cs}$ definition described in detail e.g., in [7], originally suggested in [8]. Briefly summarized, the assessment is based on the relation $E/E_c = (J_{op}/J_c)^{n(Jc)}$, where E is the electric field generated in the superconductor, J_{op} is the operating current density,

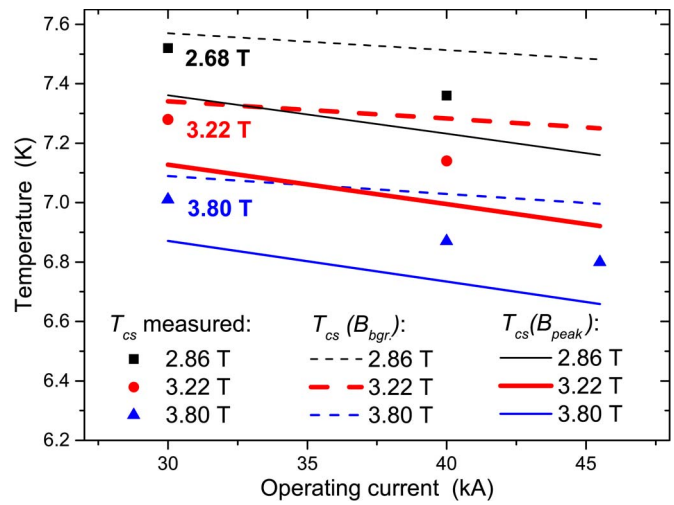

Fig. 4. $T_{\rm cs}$ as a function of operating current at different magnetic fields. The measured MBCN4 data for the right conductor section are compared to the simple predictions, in which the magnetic field across the MB conductor is constant and equal to the SULTAN background field $B_{\rm bgr.}$ (dashed lines) or to the peak field $B_{\rm peak}$. The numerical values of 2.86, 3.22, and 3.80 T correspond to the background field in all cases.

and E_c is the critical electric field set to 0.1 μV/cm, which is the value used in the experimental-data assessment. Since J_c depends on magnetic field, which varies across the cable cross section (Fig. 3), the above relation can only be applied locally at any position in the superconducting cable. The generalized equation that allows us to calculate the electric field averaged over the cable cross section, E_{av}, is

$$\frac{E_{av}}{E_c} = \frac{1}{A_{SC}} \int_{SC} \left(\frac{J_{op}}{J_c(B,T)} \right)^{n(J_c)} dx\, dy \quad (3)$$

where the integral goes over the cable current-carrying area A_{SC}. Equation (3) can be solved iteratively, and $T_{\rm cs}$ is defined as temperature T for which E_{av} is equal to $E_c = 0.1$ μV/cm.

In both approaches, i.e., in the simple one as well as in the more sophisticated one, we assume uniform current density across the cable cross section and no current redistribution among strands.

IV. Prediction Versus Measurement

A. $T_{\rm cs}$ Measurement

The comparison of the prediction to measured data is shown in Figs. 4 and 5. The first of them compares the simple assessment based on the peak magnetic field and on the SULTAN background field to the measured data. The predictions based on peak field are too conservative and clearly underestimate the experimental values of $T_{\rm cs}$ by approx. 0.15–0.2 K, while the prediction based on the background field overestimates them. The dependence of the latter prediction on the operating current is clearly different than that in the data.

The assessment based on the averaged electric field calculation defined in (3) is shown in Fig. 5 by the full lines. The agreement with the measured $T_{\rm cs}$ is better than that for the simple approach. The slope of the calculated prediction is close to slope of the measurement.

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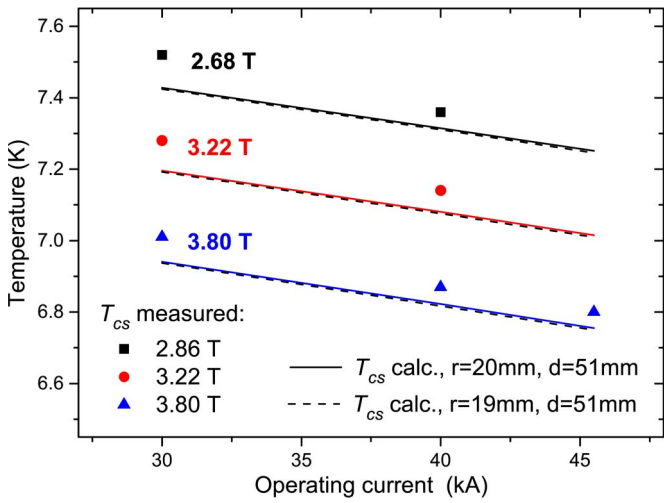

Fig. 5. $T_{\rm cs}$ as a function of operating current at different SULTAN background fields. The measured MBCN4 data for the right conductor section are compared to the prediction based on the average electric field calculation (3). The $T_{\rm cs}$ prediction for the nominal inner cable radius of $r = 20$ mm (full line) is very close to the $T_{\rm cs}$ prediction for $r = 19$ mm (dashed line). The calculations for $r = 19$ mm and conductor distance $d = 53$ mm (not shown) coincide with the prediction for $r = 20$ mm and $d = 51$ mm.

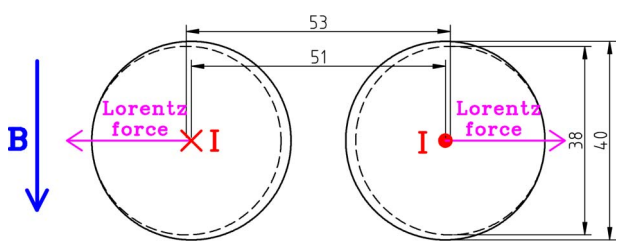

Fig. 6. Sketch of the MBCN4 sample cross section. The outer diameter of 40 mm corresponds to the cable dimension, i.e., without jacket, used in our calculations; the smaller diameter of 38 mm was chosen to illustrate the (small) effect of the conductor size on the theoretical predictions. The arrow of magnetic field B indicates the direction of the SULTAN background field.

B. Effect of Conductor Compaction

As discussed in Section II, the left conductor section has been compacted to the outer diameter of 43.3 mm instead of the nominal 44.5 mm. We theoretically investigated the effect of this conductor-geometry change in two steps. First we reduced the cable radius in both left and right conductor sections by 1 mm, which corresponds to a large conductor compaction (in fact, this compaction is almost two times larger than that done for the left conductor section). This negligibly reduced the predicted $T_{\rm cs}$, as illustrated in Fig. 5 by the dashed lines.

In the second step, we increased the distance between the centers of the compacted conductors from 51 mm to 53 mm, as indicated in Fig. 6. This change roughly corresponds to the situation, in which the strands in a superconducting cable are squeezed to the conduit wall due to Lorentz force acting on the strands during current cycling in the presence of magnetic field. This 2 mm change of conductor distance had again just a negligible ($\sim$10 mK) effect on $T_{\rm cs}$, which is opposite to the effect of radius reduction. In the end, both effects (radius reduction and conductor distance increase) practically compensated one another.

The described theoretical assessment of conductor compaction took into account only the geometrical effects, and completely ignored changes in helium cross-section and likely different inter-strand contact resistance. Experimentally, we observed just very slight difference between the left (compacted) and right (non-compacted) sample sections. An argument slightly in favor of higher compaction was a very slight increase of quench temperature T_q (in majority of runs the quench was initiated in the right sample section), presumably due to smaller inter-strand resistance. On the other hand, the nominally (less) compacted conductor reveals negligibly higher $T_{\rm cs}$ (by approx. 0.02 K), lower AC-losses (by 20% at $B = 2$ T, no transport current, after electromagnetic cycling) and obviously smaller pressure drop along the conductor length.

V. Conclusion

The measured MBCN4 conductor performance is very close to the theoretical prediction. The measured $T_{\rm cs}$ is in general 0.1 K above our theoretical assessment based purely on the WST strand properties and sample geometry, and can likely be attributed to current redistribution between strands, which is ignored in our calculations, or to the slightly different scaling law parameters than those reported in [2].

We therefore conclude that the performance of MBCN4 sample is as expected, and the $T_{\rm cs}$ degradation of $\sim$0.5 K observed at MBCN2 and MBCN3 samples was very likely related to the U-bend box. The experience from the SULTAN test facility indicates that U-bend terminations can be safely used for small size samples [9], however they should be avoided for large size conductors. Future samples of MB conductor will be assembled with bottom joint.

The larger conductor compaction did not have any significant effect on its DC performance—neither theoretically from the change of the sample geometry, nor experimentally.

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Disclaimer: The views and opinions expressed herein do not necessarily reflect those of the ITER Organization.

References

[1] B. Liu *et al.*, "The performance test and analysis of ITER main and correction busbar conductor," *Fusion Eng. Des.*, vol. 88, no. 11, pp. 2802–2809, Nov. 2013.

[2] I. Pong *et al.*, "Current sharing temperature of NbTi SULTAN samples compared to prediction using a single pinning mechanism parameterization for NbTi strand," *Supercond. Sci. Technol.*, vol. 25, no. 5, May 2012, Art. ID. 054011.

[3] P. Bruzzone *et al.*, "Upgrade of operating range for SULTAN test facility," *IEEE Trans. Appl. Supercond.*, vol. 12, no. 1, pp. 520–523, Mar. 2002.

[4] A. Devred *et al.*, "Challenges and status of ITER conductor production," *Supercond. Sci. Technol.*, vol. 27, no. 4, 2014, Art. ID. 044001.

[5] *Assembly procedure for ITER SULTAN samples*, ITER IDM Server, 2012.

[6] L. Bottura, "A practical fit for the critical surface of NbTi," *IEEE Trans. Appl. Supercond.*, vol. 10, no. 1, pp. 1054–1057, Mar. 2000.

[7] R. Wesche, A. Anghel, B. Stepanov, M. Vogel, and P. Bruzzone, "DC performance, AC loss and transient field stability of five medium size NbTi cable-in-conduit conductors with parametric variations," *Cryogenics*, vol. 45, no. 12, pp. 755–779, Dec. 2005.

[8] N. Mitchel, "Steady state analysis of non-uniform current distributions in cable-in-conduit conductors and comparison with experimental data," *Cryogenics*, vol. 40, no. 2, pp. 99–116, Feb. 2000.

[9] P. Bruzzone *et al.*, "Test results of a Nb3Sn cable-in-conduit conductor with variable pitch sequence," *IEEE Trans. Appl. Supercond.*, vol. 19, no. 3, pp. 1448–1451, Jun. 2009.

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